

Experiment- 2

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Aim:

To implement the **K-Means clustering algorithm** on the Iris dataset to classify flowers based on petal length and petal width without supervision. To implement **Multiple Linear Regression** and **Polynomial Regression** on the Iris dataset to model and predict relationships between flower features.

Task To Be Done:

For K-Means Clustering:

- Load the **Iris dataset** from sklearn.datasets.
- Select **petal length** and **petal width** features for clustering.
- Apply **K-Means algorithm** with n_clusters = 3 to group flowers into clusters.
- Assign cluster labels to the dataset and visualize the results using a **scatter plot**.

For Multiple Linear Regression:

- Select **sepal length** and **sepal width** as independent variables.
- Select **petal length** as the dependent variable.
- Train a **LinearRegression** model on the dataset.
- Predict petal lengths and visualize the results in a **3D scatter + surface plot**.

For Polynomial Regression:

- Select **sepal length** as the independent variable.
- Select **petal length** as the dependent variable.
- Transform the features into polynomial form (degree 2) using PolynomialFeatures.

- Train a regression model on the transformed data.
- Predict and visualize the polynomial curve with a **scatter plot** and a **fitted curve**.

CODE:

```
# 1. Import necessary
libraries import pandas as
pd import
matplotlib.pyplot as plt
from sklearn.cluster
import KMeans from
sklearn.datasets import
load_iris from
sklearn.linear_model
import LinearRegression
from
sklearn.preprocessing
import
PolynomialFeatures from
sklearn.metrics import
r2_score import numpy as
np
from mpl_toolkits.mplot3d import Axes3D

# 2. Load the Iris dataset from sklearn
iris = load_iris()
data = pd.DataFrame(data=np.c_[iris['data'], iris['target']],
columns=iris['feature_names'] + ['target'])

# 3. View first 5 rows to understand the data
print("\nFirst 5 rows of the dataset:")
print(data.head())

# K-MEANS CLUSTERING
print("\n--- K-Means Clustering ---")
# Use only Petal Length & Petal Width for clustering
X_kmeans = data[['petal length (cm)', 'petal width (cm)']]

# Apply KMeans clustering with 3 clusters
kmeans = KMeans(n_clusters=3, random_state=0, n_init=10)
kmeans.fit(X_kmeans)
```

```
# Add the cluster labels to the original data data['Cluster'] =
kmeans.labels_ print("\nCluster assignments for each flower:\n",
data['Cluster'].value_counts())
```

```
# Plot the clusters
```

```
plt.scatter(X_kmeans['petal length (cm)'], X_kmeans['petal
width (cm)'], c=data['Cluster'], cmap='viridis')
plt.xlabel('Petal Length (cm)') plt.ylabel('Petal Width (cm)')
plt.title('K-Means Clustering on Iris Dataset') plt.grid(True)
plt.show()
```

```
# Discussion
```

```
print("Discussion: The plot shows 3 distinct clusters of flowers based on
petal length and width.") print("These clusters roughly correspond to
the three Iris species: Setosa, Versicolor, and Virginica.\n")
```

```
#MULTIPLE LINEAR REGRESSION
```

```
print("--- Multiple Linear Regression ---")
```

```
# Use Sepal Length & Sepal Width to predict Petal
```

```
Length X_multi = iris.data[:, :2]
```

```
y_multi = iris.data[:, 2]
```

```
model_multi = LinearRegression()
```

```
model_multi.fit(X_multi, y_multi)
```

```
y_pred_multi = model_multi.predict(X_multi)
```

```
# R2 Score
```

```
r2_multi = r2_score(y_multi, y_pred_multi)
```

```
print(f"Multiple Linear Regression R2 Score:
{r2_multi:.4f}")
```

```
# Visualize in 3D
```

```
fig = plt.figure(figsize=(10, 6)) ax = fig.add_subplot(111,
projection='3d') ax.scatter(X_multi[:, 0], X_multi[:, 1], y_multi,
color='red', label='Actual') ax.plot_trisurf(X_multi[:, 0], X_multi[:,
1], y_pred_multi, color='blue', alpha=0.5, label='Prediction')
ax.set_xlabel('Sepal Length') ax.set_ylabel('Sepal Width')
ax.set_zlabel('Petal Length') ax.set_title('Multilinear Regression on
Iris Dataset') plt.legend()
plt.show()
```

```
# Discussion
```

```
print("Discussion: Multiple regression shows a linear relationship between  
sepal dimensions and petal length.")
```

```
print("R2 score indicates how well the model explains the variation in  
petal length.\n")
```

```
# ----- POLYNOMIAL REGRESSION -----
```

```
- print("--- Polynomial Regression (Degree 2) ---")
```

```
# Use Sepal Length to predict Petal
```

```
Length X_poly = iris.data[:,  
0].reshape(-1, 1) y_poly = iris.data[:,  
2]
```

```
# Polynomial transformation
```

```
poly =
```

```
PolynomialFeatures(degree=2)
```

```
X_poly_transformed = poly.fit_transform(X_poly)
```

```
model_poly = LinearRegression()
```

```
model_poly.fit(X_poly_transformed, y_poly)
```

```
y_pred_poly = model_poly.predict(X_poly_transformed)
```

```
# R2 Score
```

```
r2_poly = r2_score(y_poly, y_pred_poly)
```

```
print(f"Polynomial Regression R2 Score: {r2_poly:.4f}")
```

```
# Plot results
```

```
sorted_idx = X_poly[:, 0].argsort()
```

```
plt.scatter(X_poly, y_poly, color='red', label='Actual')
```

```
plt.plot(X_poly[sorted_idx], y_pred_poly[sorted_idx], color='green',  
label='Polynomial
```

```
Fit')
```

```
plt.xlabel("Sepal Length")
```

```
plt.ylabel("Petal Length")
```

```
plt.title("Polynomial Regression
```

```
(Degree 2)") plt.grid(True)
```

```
plt.legend()
```

```
plt.show()
```

```
# Discussion
```

```
print("Discussion: Polynomial regression captures non-linear patterns  
between sepal length and petal length.")
```

```
print("If R2 score is higher than multiple regression, polynomial regression  
is better for this data.\n")
```

Output:

```
[5]
First 5 rows of the dataset:
  sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1                3.5                1.4                0.2
1                4.9                3.0                1.4                0.2
2                4.7                3.2                1.3                0.2
3                4.6                3.1                1.5                0.2
4                5.0                3.6                1.4                0.2

  target
0    0.0
1    0.0
2    0.0
3    0.0
4    0.0

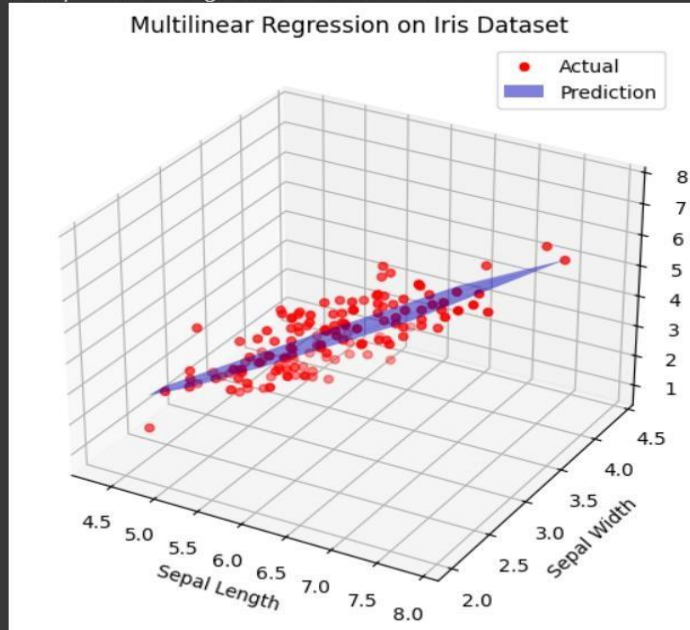
--- K-Means Clustering ---

Cluster assignments for each flower:
  Cluster
0      52
1      50
2      48
Name: count, dtype: int64
```



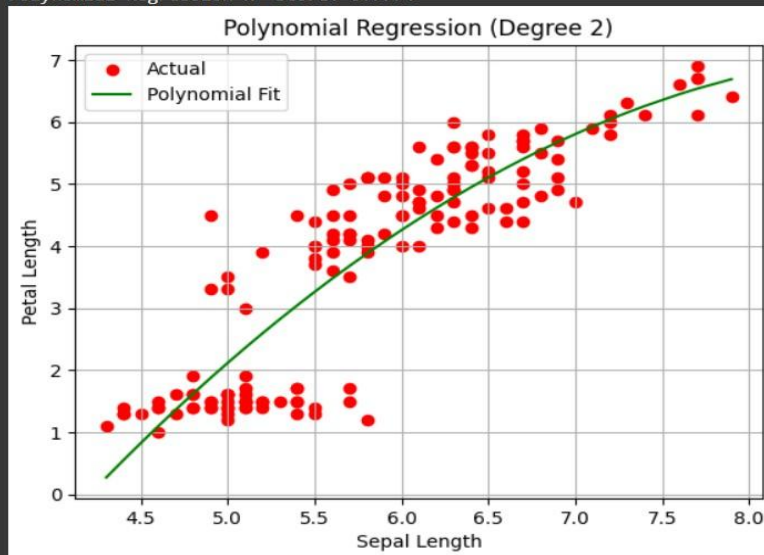
Discussion: The plot shows 3 distinct clusters of flowers based on petal length and width. These clusters roughly correspond to the three Iris species: Setosa, Versicolor, and Virginica.

--- Multiple Linear Regression ---
Multiple Linear Regression R^2 Score: 0.8677



Discussion: Multiple regression shows a linear relationship between sepal dimensions and petal length. R^2 score indicates how well the model explains the variation in petal length.

--- Polynomial Regression (Degree 2) ---
Polynomial Regression R^2 Score: 0.7774



Discussion: Polynomial regression captures non-linear patterns between sepal length and petal length. If R^2 score is higher than multiple regression, polynomial regression is better for this data.

Learning Outcomes:

- Understood the difference between **supervised** (regression) and **unsupervised** (clustering) learning techniques.
- Learned how to use **K-Means** to group data without prior labels.
- Gained knowledge of how to train and evaluate **Multiple Linear Regression** models.
- Learned how to transform features for **Polynomial Regression** to capture nonlinear patterns.
- Practiced data visualization using **Matplotlib** and **3D plots** to interpret results.